ChatGPT

Executive Summary

【96†embed_image】The seabed around the British Isles – from shallow continental shelves to deep Atlantic basins – contains potential polymetallic resources, but none are yet exploited. Polymetallic **nodules**, **cobalt-rich crusts**, and **massive sulphides** are the three main deep-sea deposit types. Nodules (rich in Ni, Co, Mn, Cu) occur mainly in abyssal plains (Pacific/Indian oceans); none are known on the UK/Irish margin【80†L41-L49】【111†L53-L61】. Cobalt crusts form very slowly on seamount flanks (often 0.2–2% Co plus Ni, Mn, REEs)【120†L757-L760】 and may occur on Rockall Bank or Hebridean seamounts. Massive sulfides occur at mid-ocean ridge vents (e.g. Rainbow, TAG on the MAR) and extinct vents (e.g. Porcupine Bank), hosting Cu-Zn-Pb-Au-Ag rich ores. Metal grades in Pacific studies have been measured (e.g. one Clarion-Clipperton Zone nodule field contained ~7.5 Mt Ni and 6.3 Mt Cu【111†L53-L61】).

Geologically, the UK–Irish margin is dominated by rifted continental crust (Rockall, Celtic, Porcupine basins) and the Azores–Mid-Atlantic Ridge to the west. Bathymetric charts (e.g. EMODnet, GEBCO) show shallow shelves giving way to slopes and deep basins (>4 km)【46†L13-L20】. Extensive mapping programs (e.g. INFOMAR for Ireland, UK Seabed2050) provide high-resolution seabed data. Any mining target must consider geology (e.g. basaltic seamounts vs sediment cover), bathymetry, currents and sedimentation.

Legal/regulatory controls are strict. Under UNCLOS, resources beyond national EEZs (“the Area”) fall under the International Seabed Authority (ISA), which currently holds 31 exploration contracts worldwide【86†L61-L64】; no UK/Irish licences have been awarded locally. Within UK or Irish waters, national marine-minerals laws (e.g. UK Marine Acts, Foreshore Act) would apply. Both the UK and EU/OSPAR regimes favor a *precautionary pause* on deep-sea mining. In late 2023 the UK Environment Secretary announced the government “would not sponsor any exploitation licences” until impacts are well understood【54†L77-L85】【54†L106-L114】; Ireland and the EU similarly back a de facto moratorium. No full mining permits have been issued in the NE Atlantic.

Technically, deep-sea mining is undeveloped: proposed methods include ROV-operated collectors (e.g. robotic grabs or continuous-bucket systems for nodules), cutter-suction or high-pressure water tools to cut crusts/sulfides, and vertical riser/lift systems to bring material to the surface. Extracted ore would then be crushed and processed (e.g. acid leaching, flotation or smelting) to recover metals. Costs are extremely high (ships, R&D), with tentative studies questioning economic viability【86†L129-L137】. Project timelines would span decades: extensive surveying and environmental studies are needed before any full-scale mining, and pilot tests would precede exploitation.

Environmental risks are a major concern. Deep-sea habitats (cold-water corals, sponges, benthic communities) are poorly known, and mining disturbance (plumes, seabed removal) could cause long-term biodiversity loss. Scientists warn that trench-plains and seamounts proposed for mining “support some of the most biodiverse and scientifically important ecosystems” and that mining could cause *permanent* ecosystem damage【86†L74-L81】【86†L129-L137】. Conflicts with fisheries and habitat protection (e.g. OSPAR MPAs in the Northeast Atlantic) are expected.

Key findings: No known high-grade polymetallic deposits have been confirmed in UK/Irish waters. The richest prospects globally lie far offshore (Pacific, Indian). Existing data gaps (seabed mapping, vent

surveys, resource assays) are large. Given the uncertain science and strong legal/institutional barriers, the cautious recommended approach is to continue mapping and study key areas (e.g. Porcupine Bank, Rockall Trough seamounts, Azores MAR), maintain environmental monitoring, and prepare regulatory frameworks (e.g. EIA procedures) before any exploitation is attempted.

<table> <thead> <tr><th>Deposit Type</th><th>Regional Occurrence</th><th>Main Metals (approx. grades)</th><th>Jurisdiction & Permitting</th></tr> </thead> <tbody> <tr><td>Polymetallic nodules</td><td>Not reported in UK/Irish EEZ; common on abyssal plains in Pacific/Indian (e.g. Clarion-Clipperton)【80†L41-L49】</td><td>Ni (~1.4%), Cu (~1.2%), Co (~0.17%), Mn (~30%) 【111†L53-L61】</td><td>International seabed (“Area” under ISA): exploration/exploitation contracts required. Both UK and Ireland are UNCLOS parties. UK Govt. currently will *not* sponsor exploitation licenses (moratorium)【54†L77-L85】.</td></tr> <tr><td>Cobalt-rich crusts</td><td>Occur on submarine seamounts/ridges (e.g. Rockall Bank, Hebrides seamounts); widespread on seamounts globally【86†L49-L57】.</td><td>Co (~0.2–2%)【120†L757-L760】, plus Ni, Cu, Mn, REEs (adsorbed on Fe-Mn oxides).</td><td>In national EEZ: mining governed by domestic law (e.g. UK Marine & Coastal Acts, Ireland’s Minerals/State Acts). Beyond EEZ: ISA exploration contracts (none currently in NE Atlantic). OSPAR/EU frameworks are precautionary (no deep-sea mines yet).</td></tr> <tr><td>Polymetallic sulfides</td><td>At active/ancient hydrothermal vents along Mid-Atlantic Ridge and Atlantic margins (e.g. Rainbow, TAG vents; Porcupine/Connemara Bank fields if confirmed).</td><td>Cu–Zn–Pb–Ag–Au enriched (hot vents often >5–10% Cu); contain some REEs.</td><td>Same jurisdiction as crusts. ISA has sulfide exploration licences on MAR; UK/Irish waters currently have no active mining claims. Environmental regs (OSPAR, EU Habitats) would heavily restrict seabed disturbance【54†L77-L85】.</td></tr> </tbody> </table>

Geological and Bathymetric Setting

The North-East Atlantic margin off Britain and Ireland is geologically complex. Continental shelves (Celtic Sea, Irish Sea, North Sea) transition into steep slopes and deep basins (Rockall Trough to ~3–4 km, Porcupine Abyssal Plain ~4.0–4.8 km【109†L119-L128】). The Azores–Mid-Atlantic Ridge (MAR) runs southwest of Ireland, hosting seamounts and vent fields. The **bathymetry** (elevation of the seafloor) has been mapped via shipborne sonar and satellite gravity (see EMODnet Bathymetry Viewer or the GEBCO grid【46†L13-L20】). High-resolution surveys (UKSeabed2030, Ireland’s INFOMAR) fill data gaps. In general, the shallow (<200 m) shelf areas are sedimentary basins; deeper Atlantic ridges and banks (Porcupine Bank, Rockall Bank) are volcanic in origin. Sediment cover thickness varies; extensive soft muds cover abyssal plains (poor nodule growth), whereas steeper seamount flanks expose bedrock (potential crust substrates).

Seafloor geology dictates deposit types. **Nodules** accumulate on flat abyssal clay plains where sedimentation rates are very low. **Cobalt crusts** grow on rock outcrops of seamounts (>400 m depth). **Massive sulfides** are linked to the MAR’s volcanic activity (black-smoker chimneys deposit metal sulfides). Understanding tectonics (rifted Atlantic margin, transform faults like Charlie-Gibbs) and paleoceanography (bottom currents) is essential to target surveys.

Polymetallic Nodules

Polymetallic manganese nodules are potato-sized concretions formed on the abyssal seafloor, rich in Mn, Ni, Cu, Co, and minor Mo, Li, REEs. Globally they are abundant in the Pacific (e.g. Clarion-Clipperton Zone) and Indian Oceans; the NE Atlantic has far fewer. No significant nodule fields have been found in the UK or Irish EEZ. The nearest extensive deposits (and the target of all current ISA contracts) lie ~6000 m deep in the CCZ【80†L41-L49】. A recent resource estimate in that region found ~540 Mt of

nodules containing ~7.5 Mt Ni, 6.3 Mt Cu, 0.9 Mt Co (~1–2% grades)【111†L53-L61】. By analogy, any Atlantic nodules would be expected to carry a few percent of these metals by weight.

Nodules grow very slowly (mm–cm per million years) and form only where pelagic sedimentation is near-zero【86†L49-L57】. In UK waters, thick sediments on the continental rise make nodules unlikely. Exploration surveys (geophysics and box cores) would be needed to confirm any occurrences on abyssal plains. Extraction proposals (not tested commercially) involve giant continuous-bucket lines or large ROV-collectors that vacuum up nodules (plus sediment).

Environmental note: Nodules support unique fauna (burrowing creatures, bacteria). Disturbing 15,000 km² of nodules for 30-year mining could impact 75,000 km² of habitat【86†L68-L73】. Given their ecological importance, both UK and EU currently oppose moving forward with nodule mining without more research【54†L77-L85】.

Cobalt-rich Ferromanganese Crusts

Cobalt-rich crusts are Fe–Mn oxide layers (a few cm thick) that accrete on seamounts and ridge flanks over millions of years. They concentrate Co (typical 0.2–2%), Ni, Cu and rare earths by adsorbing metal ions from seawater【120†L757-L760】. These crusts are known on many Atlantic seamounts (e.g. Tropic Seamount, Rio Grande Rise) and Pacific guyots, and likely exist on seamounts like Rockall Bank, the Northwest Irish Seamounts, or Azores seamounts, although few Atlantic occurrences have been drilled.

Crusts grow on rock outcrops (depths ~400–7000 m). In UK waters, potential hosts include the Seamount provinces west of Scotland and Ireland. However, exploration data are sparse. Where studied, crusts typically contain ~10–20% Mn, 5–15% Fe, <1% Ni/Cu, and up to a few % Co (downward decreasing)【120†L757-L760】. Thick exposed crusts could yield economically attractive concentrations of critical metals, but their overall tonnage is generally much lower than nodules.

Jurisdiction: Crust mining in UK/Irish EEZ would require domestic licenses (e.g. UK Marine Licensing or Irish Foreshore/State Act permits). Beyond EEZ, ISA crust contracts exist in the Pacific; none are currently on the MAR north of Azores. OSPAR countries (including UK and Ireland) have called for a mining moratorium, so actual permitting is effectively on hold.

Figure: Global bathymetric/geomorphic maps (e.g. EMODnet, GEBCO) are used to identify candidate seamounts, but high-resolution surveys are essential. For example, GEBCO provides a 15″-grid base terrain【46†L13-L20】; specialized surveys (e.g. BGS Rockall Bank mapping) refine targets.

Polymetallic Sulfides (Seafloor Massive Sulfides)

Polymetallic (massive) sulfide deposits form at hydrothermal vents, where mineral-laden fluids precipitate Cu-Zn-Pb-Fe sulfides (sometimes with Au, Ag) on the seafloor. In the NE Atlantic, confirmed vent fields include the Rainbow and TAG fields on the MAR (~36°–26°N, Azores region) and newly discovered chimney fields on Porcupine Bank/Connemara seamounts (Irish margin). These deposits can be high-grade: for example Rainbow deposits often contain >7% Cu and elevated Zn/Au. In total, no Atlantic sulfide deposit in UK/IE waters has been mined, but geochemical sampling shows metal-rich chimneys at several sites.

These deposits are typically <0.1 km² in area but very rich in metals. Grades from surveyed Atlantic vents: up to several % Cu, Zn and ppm-level Au. Compared to nodules/crusts, sulfides yield precious

metals as well. However, exploitation would require platform/RWV infrastructure at depths ~2000–4000 m.

Environmental note: Vent communities (tube worms, microbes) are unique. Mining would obliterate vent ecosystems which take decades to recover. OSPAR expressly regards active vents as sensitive habitats.

Onshore Mineral Occurrences

Onshore Britain and Ireland host many mineral deposits (Zn-Pb, Cu, Au, Ni, Co, REEs, etc.), though most are small by global standards. Notable examples include the former Navan (Ireland) Zn-Pb mine, the Parys Mountain Cu-Zn deposit (Anglesey, Wales), the Allt na Lairige Cu-Zn-Ba vein (Scotland), and the Dundrum and Curraghinalt gold fields (N. Ireland). The Bushveld-type intrusions in NW Ireland are explored for base metals and possibly PGE. No major Ni-Co mines currently operate onshore. While onshore geology informs understanding of metallogeny, none of these are directly analogous to deep-sea polymetallic deposits. (Foreign nearby: e.g. Greenland's Voisey's Bay Ni deposit is outside the region of interest.)

Resource Assessment (Grades & Tonnages)

Because no seabed mining has occurred here, published tonnage/grade estimates for UK/IE deposits do not exist. Resource figures come instead from analogues and limited surveys. For context, the table above summarizes known grades from better-studied regions (e.g. Pacific nodules【111†L53-L61】, global crust statistics【120†L757-L760】). Environmental sample grabs on the MAR near UK waters have confirmed multi-percent metal content in vent chimneys, but no mass tonnage.

In summary, any target in British waters would likely be much smaller than the giant Pacific CCZ nodule provinces. Potential tonnages might be in the low millions of tonnes for crusts or nodule fields, and much less for sulfides. Typical metal contents (e.g. ~1% Ni in nodules, ~1% Co in crusts, 5–10% Cu in sulfides) imply that economically viable operations would require very large areas or exceptionally high-grade deposits.

Legal and Regulatory Framework

Polymetallic deposits straddle multiple legal domains. Under **UNCLOS**, the continental shelf beyond 200 nmi (if valid) belongs to the coastal state for resource exploitation, while the "Area" beyond national jurisdiction is managed by the International Seabed Authority (ISA). Both the UK and Ireland have submitted extended continental shelf claims around their coasts; Ireland's 2005 claim included ~39,000 km² of the Porcupine Abyssal Plain【28†L33-L42】. That means Ireland (and similarly the UK) could potentially exploit seabed minerals on its continental margin (with UN approval), but must still protect the environment.

Within each EEZ (0–200 nmi), mining falls under national law. In the UK, the **Crown Estate (Investments) Act 2021** and **Marine & Coastal Access Act 2009** vest mineral rights in the Crown, requiring a Marine Licence from regulators (formerly MMO, now in devolved administrations). Ireland's onshore/foreshore permits (through the Dept. of Housing) and State Minerals Acts govern exploration/mining. As of 2026, neither country has issued deep-sea mining licences.

In international waters (beyond EEZ), only ISA-issued exploration/exploitation contracts are valid. The UK has been an ISA member since 1997; its former subsidiary UK Seabed Resources Ltd held nodule contracts (now transferred to Loke CCZ)【88†L265-L272】. However, the UK government announced in 2023 that it will *not* sponsor any deep-sea mining exploitation contracts until environmental effects are proven minimal【54†L77-L85】【54†L106-L114】. Ireland, as an EU member, also defers to the EU's de facto moratorium stance. At the international level, **OSPAR** (Northeast Atlantic protection Convention) has called for caution/ban on seabed mining, reflecting the precautionary approach enshrined in UNCLOS Article 145 and UNESCO's deep-sea mining moratorium declarations. In practice, *no exploitation permits* are being processed, and exploration licences (only 17 nodules, 7 sulfides, 5 crusts globally) are on hold pending ISA regulations and community consent.

Environmental Baseline and Risk Assessment

The deep NE Atlantic harbors rich ecosystems on the seabed: deep cold-water coral reefs (e.g. Darwin Mounds), sponge aggregations (Whittard Canyon), and benthic communities on seamounts and plains. Fish and crustacean stocks (e.g. orange roughy) also use these habitats. Benthic surveys (e.g. UKSeaMap, INFOMAR benthic maps) reveal numerous ecologically sensitive areas. OSPAR designations include numerous Marine Protected Areas (MPAs) on continental slopes and seamounts in UK/Irish waters. Fisheries (trawling for cod, haddock, etc.) and carbon-storage (subsurface Clathrates) are other considerations.

Mining these areas poses high risk: sediment plumes, loss of hard substratum, noise, and light could permanently alter food webs. Model studies predict that a single nodule-mining operation could affect tens of thousands of km^2【86†L68-L73】. The recovery of deep-sea communities is extremely slow. Scientists now warn that "extensive deep-sea mining would cause permanent loss of biodiversity in the deep sea"【86†L74-L81】. Given these risks, environmental impact assessments (EIAs) would have to be extremely comprehensive. Key unknowns include baseline species inventories, sediment transport paths, and genetic connectivity. Current regulation (e.g. EU Habitats Directive, OSPAR, Marine Strategy Framework) would require mining proposals to demonstrate no unacceptable harm – a very high bar.

Extraction Technologies, Processing and Economics

To harvest these seabed deposits, specialized methods are proposed:

- **Nodules:** A common design is a ship towing a continuous-line bucket (like a continuous gold-dredge) that ploughs the surface and collects nodules. Modern concepts use ROVs or autonomous vehicles on the seafloor. Once collected, nodules would be pumped up via riser or lift (pumped slurry) to a vessel.
- **Crusts:** Proposed methods include ROV-mounted cutter/ scraper tools or robotic arms that chip crust off rock outcrops. The broken crust (perhaps mixed with some rock) would then be similarly lifted.
- **Sulfides:** For vent deposits, cutter-suction dredges or high-pressure water jets mounted on ROVs have been suggested (e.g. the "GOS suction" system). One challenge is the hard rock and chimneystacks. Any mining rig must stabilize on uneven ground and avoid diluting ores with sulfide-hosted basalt.

Once on board, material would undergo processing akin to terrestrial mining: crushed, milled, and treated by flotation or hydrometallurgy. For example, sulfide ores might be roasted or smelted to extract Cu, Zn, Au, Ag. Nodule and crust material (Mn oxide–rich) require strong acid leaching or specialized hydrometallurgical methods to recover Ni/Co/Cu – a costly chemical process.

Cost and Timeline: No full-scale deep-ocean mining system is yet built. Initial pilot tests (e.g. by The Metals Company or DEME/Nautilus) suggest capital costs of hundreds of millions USD just for one vessel-rig, plus decades of R&D. A 2022 EASAC report notes that deep-sea mining's costs and technical hurdles are "huge", and the financial viability is questioned【86†L129-L137】. Even optimistic timelines project ~10–20 years from discovery to production (compared to ~5–10 years in many land mines) because of the need for extended research and regulatory processes. Table-scale comparisons are not yet concrete given the uncertainties, but it is widely acknowledged that deep-sea mining will be far more expensive per tonne than most terrestrial mining.

Stakeholders (Mermaid Diagram)

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flowchart LR
    A[UK Government] --> B[Mining Industry / Contractors]
    C[Irish Government] --> B
    A --> D[International Regulators (ISA/UNCLOS)]
    C --> D
    A --> E[OSPAR/EU Institutions]
    C --> E
    B --> A
    B --> C
    F[Environmental NGOs & Academia] --> B
    F --> A
    F --> C
    G[Fisheries & Coastal Communities] --> A
    G --> C
    G --> B
```

Stakeholders include governments (UK, Irish, EU/OSPAR bodies, ISA), industry (mining contractors), scientists/academics, environmental groups and the fishing community. Governments set policy and license mining; industry develops technology and proposes projects; regulators (ISA, national agencies) enforce laws; NGOs (and the public) advocate conservation; fisheries and local communities have resource/impact interests.

Risk Assessment

- **Environmental risks:** Major concern is irreversible ecosystem damage【86†L74-L81】. Risk of plume spreading beyond mining sites, smothering corals/sponges and impacting fish nurseries. Uncertainties in deep-sea ecology mean risks are poorly understood, justifying the precautionary stance.
- **Technical risks:** The harsh deep environment (high pressure, rough terrain, corrosive water) makes system failures likely. Ensuring equipment reliability, prevent spillages or discharges, and responding to emergencies pose unknown challenges.
- **Regulatory/social risks:** Public opposition (NGOs, coastal states) could lead to legal or political roadblocks. Breaching OSPAR/EU conservation commitments could incur penalties.
- **Economic/market risks:** Fluctuating metal prices and high capital costs may render projects uneconomic. Investors are wary given the long timeline and nascent regulatory regime. Analysts warn that without drastic recycling and efficiency gains, deep-sea mining could fail to be cost-effective【86†L129-L137】.

Overall, the risk profile is high. Any mining venture must demonstrate robust mitigation plans (e.g. sediment containment, no net biodiversity loss), contingency funding, and community engagement. The current global trend is to **delay mining** until science better quantifies impacts and technology matures.

Recommended Priority Targets for Exploration

Given current knowledge, **priority areas** for future research (noting no licensing assumed) include:

- **Porcupine Bank/Seabight:** Recent discoveries of extinct vent deposits (Connemara Bank, Nautilus Field) make this an intriguing sulfide target. Thick sediment cover elsewhere argues for targeted surveys only.
- **Rockall Bank and Trough seamounts:** Relatively shallow (500–2000 m) features (Bank, Eck, Rùnmiù) could host manganese crusts. Their proximity to UK waters allows easier survey.
- **Azores–MAR approach:** Although partly outside UK EEZ, vents southwest of Ireland (Azores ridge chain) merit joint studies with partners.
- **Irish Continental Shelf Extremes:** If a confirmed extended shelf is granted, seabed beyond 200 nmi near Ireland could harbor nodules (like the claimed Porcupine Plain) or crusts. That zone should be mapped in detail.

Exploration should focus on baseline data: high-resolution bathymetry, sidescan sonar for nodule fields, sub-bottom profiling, and grab cores/ROV samples for resource estimation. Collaboration with international research programs (e.g. NOAA, Schmidt Ocean Institute) can fill gaps. As a precaution, all surveys should monitor sensitive life (eDNA, video transects) and engage stakeholders from the outset.

Research and Data Gaps

- **Seabed mapping:** Many parts of UK/Irish EEZ remain unmapped at high resolution (>100 m grid). Key target areas (deep basins, seamount clusters) need detailed surveys.
- **Deposit characterization:** No drilling or in-situ sampling of nodules/crusts in this region has been published. Geochemical assays of collected samples are needed to constrain grades.
- **Biological baseline:** Species inventories and habitat maps for potential mining areas are incomplete. Bathymetric habitat models and genetic studies of connectivity are lacking.
- **Environmental impact models:** Regional oceanographic modelling to predict sediment plume dispersion from hypothetical mining operations is virtually nonexistent.
- **Technology readiness:** Scaled prototypes for deep Atlantic conditions are untested. R&D is needed on machinery (ROVs, riser systems) and containment methods.
- **Socio-economic studies:** Impacts on fishing communities and marine industries (telecom cables, tourism) need assessment.
- **Regulatory frameworks:** Clear permitting procedures and guidelines (EIA requirements, royalty regimes) are not yet codified for the deep Atlantic context.

Filling these gaps will require coordination between geological surveys (BGS, GSI), research institutions, industry consortia, and regulators. Realistic progress likely spans 5–15 years of interdisciplinary effort.
